

Ashutosh Joshi (H1-B Visa)

Software Engineer | Maryland, United States | http.ashutosh@gmail.com | +1-551 344 6092 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Philomath Data Scientist and Software Engineer with 4+ years of experience delivering end-to-end analytics—from cloud ETL and Big Data processing to machine learning and BI storytelling. Partners with NASA/ESA at SciGlob on Pandonia Global Network, leveraging insights from 300+ spectroscopic units worldwide to support atmospheric research and data-driven decisions.

PROFESSIONAL EXPERIENCE

Software Engineer- NASA, GSFC

SciGlob Instruments & Services, Columbia MD | Nov 2024 – Present

- Built an automated data acquisition + cleaning pipeline aggregating atmospheric measurements from 300+ globally deployed Pandora units, delivering analysis-ready datasets for researchers and scientists.
- Developed a status dashboard web app to monitor 300+ units end-to-end (device health, data-fetching, logging, and operational states), improving visibility and faster issue triage across the network.
- Packaged and deployed cross-platform internal tools by shipping Windows executables and macOS apps for lab/field systems (e.g., NO₂ sonde, O₃ sonde), enabling reliable one-click installs on in-house machines.
- Designed and implemented ML + deep learning anomaly detection at scale across 300+ units to automatically flag sensor/data irregularities and reduce manual monitoring burden.
- Built an encoder-decoder CNN for cloud detection and integrated it into the Sun-scan measurement workflow to condition/trigger measurements based on sky conditions for higher-quality scans.
- Engineered robust, user-friendly hardware control software to operate critical instrument subsystems (e.g., AC motors, Oriental high-functional motors, rain sensors, spectrometers, lasers) with safe command sequencing and fault-tolerant behavior.
- Created multiple data visualization dashboards to translate complex atmospheric datasets into clear, decision-ready insights for scientists and stakeholders.

Data Analyst & Developer

407 Associates LLC, Laurel, MD | Apr 2024 – Nov 2024

- Built clean, reliable datasets for analytics and ML by transforming raw data, handling missing values/outliers, and standardizing inputs so teams could trust the numbers.
- Partnered with product and engineering to deliver predictive insights—developing scalable Python ML pipelines (Scikit-learn, PySpark, TensorFlow) and improving operational decision-making.
- Managed cloud data storage and performance (AWS S3, Snowflake) and wrote complex SQL to make data faster to access, easier to analyze, and cheaper to run.
- Turned complex analysis into clear business actions through Tableau storytelling and interactive Power BI dashboards—driving data-driven decisions and reducing costs by 25%.

System Engineer & Assistant System Engineer

Tata Consultancy Services, Bangalore, IN | Jun 2020 – Aug 2022

- Cleaned and prepared 10M+ rows of data across multiple sources for analysis and downstream pipelines, improving dataset readiness and reducing rework.
- Optimized data integration workflows by adding 20+ automated validation checks and refining transformation logic, cutting data-quality defects by 30% and improving consistency across feeds.
- Delivered 10–15 stakeholder-ready reports/dashboards in Python + Power BI, translating complex integration results into clear insights for design and implementation decisions.
- Defined and tracked 15+ key metrics (accuracy, freshness, latency, completeness) and presented weekly findings to project teams, helping improve decision-making speed by 20% and strengthening requirements-to-data-flow alignment.
- Improved cloud application reliability by running performance tests and tuning bottlenecks, raising throughput by 25%, reducing response time by 30%, and supporting 99.5%–99.9% availability targets through monitoring and automation.
- Streamlined DevOps delivery by building deployment/monitoring scripts and contributing to CI/CD pipelines across 3–5 environments, reducing release time by 40% and lowering incident MTTR by 25% through faster debugging and clearer runbooks/documentation.

TECHNICAL SKILLS

Programming & Development: Python, Java, R, SQL, PySpark, Scala, Flask, ReactJS, REST APIs

Data Engineering & ETL: ETL pipelines, data modeling (schemas, normalization), Apache Spark, Hadoop, AWS Glue, Snowflake

Data Analysis & Statistics: EDA, trend/pattern analysis (Pandas, NumPy, Excel), hypothesis testing, confidence intervals

Databases & Query Optimization: SQL (joins, window functions), query optimization, AWS Redshift, AWS S3

BI, Visualization & Delivery: Tableau, Power BI (interactive dashboards/reports), Matplotlib, Seaborn, Plotly, Git/GitHub, Jenkins, Agile/Scrum, JIRA, Kanban, Cloud platforms (AWS, Azure, GCP)

KEY PROJECTS

 [GitHub](#)

EM27 – Bruker

- Python-first GUI app for the EM27 instrument stack, with the repo primarily implemented in Python, indicating a desktop UI + logic split.
- Modular architecture: separates business/device logic into dedicated `controllers/` and `drivers/` folders, making the codebase easier to extend and maintain.
- Clear entry points for running the application, centered around `main_gui.py` (GUI layer) and `run_app.py` (launcher/runtime script).
- Multi-language support where needed suggesting integration with lower-level components or performance/device interfaces.

SciGlobV4.0

- It is an operator-friendly desktop app that runs the whole measurement workflow—it resets and tracks motor angle, dynamically adjusts spectrometer integration time based on signal intensity, tracks filter-wheel position, and writes these states into CSV logs so each scan is traceable.
- Built in a clean, modular way (controllers/drivers/gui) with hardware configured via JSON, and includes required instrument dependencies (e.g., spectrometer DLL) so the same software can be set up consistently on lab machines.
- It's designed for day-to-day usability in the lab/field—with UI improvements like a square camera view and branded header, plus an outlined path to add THP (temperature/humidity/pressure) live plots and logging in the same interface.

EDUCATION

University of Maryland Baltimore County

Masters of Professional Studies in Data Science

GPA: **3.89 / 4.00** | Concentration: Data Science/Machine Learning

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Computer Organization & Programming, Neural Networks, Machine Learning, Statistics & Applications, Big Data & Hadoop

Baltimore, MD

Aug 2022 - May 2024